For a scheme $[\![\mathrm{XYZYX}]\!]$, we assume that each of the X, Y and Z characters in the string are distinct, and correspond to one of the A, B or O pieces of our splitting strategy. Thus we have $\mathcal{L}_0^\dagger = \mathcal{L}_{\mathrm{LD}}^\dagger$.

Evidently, the scheme's associated Fokker-Planck operator $\hat{\mathcal{L}}^\dagger$ no longer has a leading order term that is linear in the stepsize, so it makes sense to revise our assumption on the form of the scheme's corresponding invariant measure, and replace it with a series with solely even order terms. If we suppose that the odd order correction functions are zero, then we are left with

$$\hat{\rho} = \rho_\beta(1 + h^2 f_2 + h^4 f_4 + \ldots), \tag{7.27}$$

where each $\langle f_i \rangle = 0$ as usual. Thus the stationarity equation $\hat{\mathcal{L}}^\dagger \hat{\rho} = 0$ becomes

$$\left(\mathcal{L}_{\mathrm{LD}}^\dagger + h^2 \mathcal{L}_2^\dagger + \ldots\right)\left(\rho_\beta + h^2 \rho_\beta f_2 + \ldots\right) = 0.$$

The equation is satisfied at zeroth order, and equating the second order terms we obtain

$$\mathcal{L}_{\mathrm{LD}}^\dagger\left(\rho_\beta f_2\right) = -\mathcal{L}_2^\dagger \rho_\beta. \tag{7.28}$$

The procedure then follows just as in the case of the first order schemes. The form of the right-hand side $\mathcal{L}_2^\dagger \rho_\beta$ will be dependent on the choice of the update scheme, and can be computed using the BCH formula using Proposition 7.1. More complicated symmetric schemes involving more than five stages can be utilized, but these schemes may require more than one force evaluation per iteration, while increasing the complexity of the PDE to solve.

The condition on the existence of such a solution requires us to use the Fredholm alternative, which requires us to verify that the inhomogeneous term $\mathcal{L}_2^\dagger \rho_\beta$ integrates to zero.

7.9.1 Calculation of the Inhomogeneous Term

The computation of the right hand side in (7.28) is required in order to solve the corresponding partial differential equation and obtain the leading order correction function for our symmetric scheme $[\![\mathrm{XYZYX}]\!]$. From the form of the $\mathcal{L}_2^\dagger$ operator in Proposition 7.1, we can see that our life is made considerably easier if we initially consider methods $[\![\mathrm{OYZYO}]\!]$, which substitutes

$$\mathcal{L}_X^\dagger = \mathcal{L}_{\mathrm{O}}^\dagger, \qquad \mathcal{L}_Y^\dagger + \mathcal{L}_Z^\dagger = \mathcal{L}_{\mathrm{A}}^\dagger + \mathcal{L}_{\mathrm{B}}^\dagger = \mathcal{L}_H^\dagger,$$

simplifying the computation to

$$\mathcal{L}^{\dagger}_{2,[\![\mathrm{OYZYO}]\!]}\rho_\beta = \frac{1}{12}\left[\mathcal{L}^{\dagger}_{\mathrm{Z}},\mathcal{L}^{\dagger}_{\mathrm{Z}},\mathcal{L}^{\dagger}_{\mathrm{Y}}\right]\rho_\beta - \frac{1}{24}[\mathcal{L}^{\dagger}_{\mathrm{Y}},\mathcal{L}^{\dagger}_{\mathrm{Y}},\mathcal{L}^{\dagger}_{\mathrm{Z}}]\rho_\beta. \tag{7.29}$$

The inhomogeneity for the schemes $[\![\mathrm{OABAO}]\!]$ and $[\![\mathrm{OBABO}]\!]$ is hence a combination of the two commutator terms

$$\begin{aligned}\left[\mathcal{L}^{\dagger}_{\mathrm{A}},\mathcal{L}^{\dagger}_{\mathrm{A}},\mathcal{L}^{\dagger}_{\mathrm{B}}\right]\rho_\beta &= \left(2\beta \boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)\rho_\beta,\\ \left[\mathcal{L}^{\dagger}_{\mathrm{B}},\mathcal{L}^{\dagger}_{\mathrm{B}},\mathcal{L}^{\dagger}_{\mathrm{A}}\right]\rho_\beta &= -\left(2\beta \boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U\right)\rho_\beta,\end{aligned}$$

giving overall equations to solve

$$\begin{aligned}\mathcal{L}^{\dagger}_{\mathrm{LD}}\left(\rho_\beta f_{2,[\![\mathrm{OABAO}]\!]}\right) &= -\mathcal{L}^{\dagger}_{2,[\![\mathrm{OABAO}]\!]}\rho_\beta\\ &= \frac{\beta}{24}\left(6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)\rho_\beta,\\ \mathcal{L}^{\dagger}_{\mathrm{LD}}\left(\rho_\beta f_{2,[\![\mathrm{OBABO}]\!]}\right) &= -\mathcal{L}^{\dagger}_{2,[\![\mathrm{OBABO}]\!]}\rho_\beta\\ &= \frac{\beta}{24}\left(2\nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p} - 6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U\right)\rho_\beta.\end{aligned}$$

We can see immediately that the Fredholm alternative holds in both cases, as both equations have right-hand sides that are of odd order in p, and hence will average to 0.

If we instead consider schemes which use a central Ornstein-Uhlenbeck update, then the inhomogeneous term becomes significantly more complex, as we no longer have a significant number of cancellations in $\mathcal{L}^{\dagger}_2\rho_\beta$. Instead, we shall avoid computing the right-hand side for these schemes altogether, in favor of utilizing the following lemma [223]:

Lemma 7.1 *Consider two numerical schemes with associated operators governing the evolution of measure $\mathcal{ST}$ and $\mathcal{TS}$, with unique associated invariant distributions $\hat{\rho}_{ST}$ and $\hat{\rho}_{TS}$ respectively, such that*

$$\mathcal{ST}\hat{\rho}_{ST} = \hat{\rho}_{ST}, \qquad \mathcal{TS}\hat{\rho}_{TS} = \hat{\rho}_{TS}.$$

Under suitable ergodicity assumptions (such that in the long-time limit, all distributions converge to the invariant distribution), we have

$$\hat{\rho}_{ST} = \mathcal{S}\hat{\rho}_{TS}, \quad \textit{and} \quad \hat{\rho}_{TS} = \mathcal{T}\hat{\rho}_{ST}.$$